Diagnosis: Interface Energy Loss of the Impedance-Overlap Schwarz Coupling on Nonconforming Meshes, and an Energy-Consistent Redesign

Alejandro Mota

July 11, 2026

Abstract

A 4×3×3 parametric study of the impedance-overlap Schwarz coupling on the bent-cantilever release problem (integrator pairs II/IE/EI/EE × overlap widths $\{0.0508, 0.0254, 0.0127\}$ m × mesh ratios $\{1{:}1, 3{:}4, 1{:}2\}$, 1 ms at $\Delta t = 10^{-6}$ s) shows the partition-of-unity energy conserved to 0.1% on conforming meshes but losing 28–71% — and in one case *gaining* 12% — on nonconforming ones. A follow-up 108-run sweep over mesh ratios 1:1.0–1:0.2 (Sec. 3) shows the gain is generic, not exceptional: even with variational transfer, mild coarsening injects up to +45%, one configuration pumps the beam to +275% of its initial energy, and the sign of the drift flips between adjacent mesh ratios. Channel-resolved interface-work instrumentation and an impedance-scale sweep identify the cause: the impedance dashpot $\boldsymbol{Z}(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}})$ acts on an interface velocity jump that cannot vanish at Schwarz convergence when the two trace spaces differ, and because the two subdomains transfer partner data through *independent, non-adjoint* interpolations, the dashpot work is not even sign-definite. The optimized-Schwarz, multi-time-step, and discontinuous-Galerkin literatures converge on the same two structural requirements for an energy-consistent coupling: (i) the kinematic transfer and the force transfer between the two sides must be L^2-adjoints through a single variational cross-mass matrix, which makes the interface power telescope to a sign-definite dashpot on the common-space jump plus a *conservative* spring term; and (ii) the dashpot must act only on content both trace spaces can represent, so that it vanishes at Schwarz convergence instead of persistently absorbing unrepresentable fine-scale content. These requirements are implementable only on the *nonoverlap* impedance variant (single shared interface), where — together with a dynamically consistent d'Alembert interface traction and an IMEX treatment of the interface rows for explicit subdomains — they are now implemented and validated: implicit–implicit drift improves from −9.6% to −0.03% on conforming and mildly nonconforming meshes, the conforming explicit–explicit trajectory matches the monodomain explicit run to 0.35% pointwise, and every injection pathology disappears (Sec. 7). Under subcycling, two infrastructure defects that had silently reduced the multi-time-step exchange to end-sampled partner data are fixed — genuine Gravouil–Combescure interpolation then halves the excess dissipation at strong mesh contrast — and the Prakash–Hjelmstad exact-telescoping refinement is measured to be structurally incompatible with a partitioned exchange: its work-conjugate data are either lagged (over-dissipative) or anticipatory (unstable), so exact subcycled telescoping requires a direct interface solve (Sec. 7.4). This note records the evidence, the theory, the phased redesign, and the validated fixes.

1 Observed behavior

The benchmark is the bent 3D cantilever released from a parabolic initial displacement, split lengthwise into clamped and free subdomains that overlap, coupled by the impedance-overlap transmission condition

$$\boldsymbol{t} \;=\; \boldsymbol{t}_p \;+\; \boldsymbol{Z}\,(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}}) \;+\; \alpha\,(\boldsymbol{u}_p - \boldsymbol{u}), \qquad \boldsymbol{Z} = Z_p\,\boldsymbol{n}\otimes\boldsymbol{n} + Z_s\,(\boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n}), \tag{1}$$

with $Z_p = \rho c_p$, $Z_s = \rho c_s$ of the partner material (Lysmer–Kuhlemeyer split), $\alpha = 0$ throughout this study, and subscript p denoting partner fields evaluated at this subdomain's Schwarz boundary. Energy is the Arlequin partition-of-unity total (overlap counted once), normalized by each run's own $E(0) \approx 1506\,\mathrm{J}$.

Drift $E(1\,\mathrm{ms})/E(0) - 1$, in percent, pointwise transfer:

overlap	ratio	II	IE	EI	EE
0.0508 (max)	1:1	−0.07	−8.9	−0.2	+8.7
	3:4	+1.1	−8.7	−0.9	−12.0
	1:2	+12.5	−26.5	−53.8	−57.7
0.0254 ($\frac{1}{2}$)	1:1	−0.07	+5.0	−0.4	+10.2
	3:4	−28.4	−6.0	−22.3	−13.9
	1:2	−17.3	−18.5	−58.9	−45.7
0.0127 ($\frac{1}{4}$)	1:1	−0.02	−15.7	−6.1	+3.6
	3:4	−44.5	−40.6	−41.1	−31.4
	1:2	−63.8	−64.3	−70.8	−61.9

Three regimes: (a) conforming (1:1) implicit-free-subdomain runs conserve to $\leq 0.1\%$ at every overlap; (b) nonconforming runs lose tens of percent, worse with mesh ratio and with smaller overlap; (c) a minority of cases, including II at max overlap 1:2, *gain* energy. Variational (`transfer: variational`) rescues the small-overlap losses dramatically ($-44.5 \to -4.4\%$ at $\frac{1}{4}$/3:4) but regresses elsewhere ($+12.5 \to -50.7\%$ at max/1:2): it improved 17 of 24 nonconforming cases, mean |drift| $33.4 \to 25.4\%$ — helpful, not structural.

2 Empirical diagnosis

2.1 Channel-resolved interface work

The instrumentation (`NORMA_IMPEDANCE_WORK_CSV`) decomposes each subdomain's interface power $P = \sum_i \dot{\boldsymbol{u}}_i \cdot \boldsymbol{f}_i^{\mathrm{bc}}$ into the traction, dashpot, and Robin channels of (1). For the exact monodomain solution the dashpot and Robin channels vanish identically, so their work is purely spurious. Integrated over 1 ms (II runs; work summed over both subdomains):

case	ΔE blended	dashpot work	traction work	jump RMS
conforming ($\frac{1}{2}$, 1:1)	−1 J	−1.8 J	+70 J	~ 1
nonconf. $\frac{1}{2}$, 3:4 (pointwise)	−428 J	−586 J	+92 J	~ 20
nonconf. $\frac{1}{4}$, 3:4 (pointwise)	−515 J	−877 J	+341 J	~ 16
nonconf. max, 1:2 (pointwise)	+188 J	**+554** J	−211 J	~ 80
nonconf. $\frac{1}{4}$, 3:4 (variational)	−67 J	−864 J	+780 J	~ 26

Three facts follow. First, on conforming meshes the dashpot channel is negligible ($-1.8\,\mathrm{J}$) and the coupling is conservative: the jump $\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}}$ closes at Schwarz convergence. Second, on nonconforming meshes the dashpot channel carries the loss: the jump RMS is an order of magnitude larger and never closes — the coarse trace space cannot represent the fine side's boundary velocity, and the dashpot persistently absorbs the difference, exactly as a Lysmer–Kuhlemeyer boundary absorbs outgoing content. Third, and decisively for the mechanism: in the max/1:2 case the "dashpot" does +554 J of *positive* work. A dissipator can never do positive work; this proves the discrete dashpot term is not a quadratic form. Because each subdomain interpolates partner fields with its own, unrelated operator, the two sides' dashpot terms do not pair, and the total is sign-indefinite.

The variational row explains why variational transfer helps only sometimes: its dashpot still drains −864 J, but the consistent (characteristic-transfer) traction returns +780 J. The rescue is an accidental near-cancellation of two large opposing channels, not a removal of either defect.

2.2 Impedance-scale sweep

Scaling both impedances by $s \in \{1, 0.5, 0.25\}$ on the $\frac{1}{4}$/3:4 II case (pointwise):

impedance scale s	1.0	0.5	0.25
drift at 1 ms	−44.5%	−40.5%	−29.2%

The loss decreases monotonically as the dashpot is weakened — confirming the dashpot as the carrier of the loss — but sublinearly: the dissipated power is $Z\,[\![\dot{u}]\!]^2$ and the jump itself grows as the absorber detunes (reflection increasing as $Z \to 0$), so the product falls more slowly than Z. This is the signature of an absorbing-boundary mechanism rather than of a fixed damping force, consistent with the channel decomposition above.

2.3 Control experiment: hard Dirichlet matching

To separate "defect of the impedance condition" from "price of nonconforming overlap coupling," the same nine mesh configurations were run with the classical Dirichlet-overlap coupling (II), in both transfer flavors — pointwise interpolation and the weak (variational L^2-projection) Dirichlet option:

overlap	ratio	DBC pointwise	DBC weak
0.0508 (max)	1:1	+0.05%	—
	3:4	+5507%*	+2418%
	1:2	+2765%*	+5752%*
0.0254 ($\frac{1}{2}$)	1:1	−0.03%	—
	3:4	+423%	+749%
	1:2	+5628%*	+7765%*
0.0127 ($\frac{1}{4}$)	1:1	+0.01%	—
	3:4	+33%	+22%
	1:2	+3268%*	+262%

* run died (Newton failure) before 1 ms; drift at last stop.

Hard matching conserves essentially exactly on conforming meshes and is *violently unstable* on every nonconforming one — and the weak (contraction) transfer does *not* rescue it. Two consequences. First, the impedance dashpot's dissipation is not merely a defect: it is what *stabilizes* a coupling that hard kinematic exchange destabilizes; its magnitude is the energy in the inter-grid disagreement it must continually absorb. Second, the Dirichlet instability grows with overlap *width* (opposite to the impedance losses), consistent with a feedback loop between the two interpolation surfaces through the overlap volume.

3 Mesh-ratio sweep: injection is generic

An independent, finer-grained parametric study by our intern Lukas reported significant energy *injection* at mesh ratios the original $\{1{:}1, 3{:}4, 1{:}2\}$ grid does not sample. The following sweep confirms and maps it: same beam, release, and symmetric subdomains $L_{\text{sub}} = (0.254 + \text{ov})/2$; free mesh fixed at $h = 6.35$ mm; clamped mesh coarsened to $6.35/r$ mm for $r = 1.0, 0.9, \ldots, 0.2$ (clamped:free element-size ratio 1:r; 1:0.1 skipped); all three overlap widths; all four integrator pairs; variational transfer (now the default) throughout — 108 runs. The 1:1.0 column reproduces the conforming baselines of Sec. 1 to all printed digits, and 1:0.5 reproduces the earlier variational 1:2 results exactly.

	overlap 0.0508 m				overlap 0.0254 m				overlap 0.0127 m			
ratio	II	IE	EI	EE	II	IE	EI	EE	II	IE	EI	EE
1:1.0	−0.1	−8.9	−0.2	+8.7	−0.1	+5.0	−0.4	+10.2	−0.0	−15.7	−6.1	+3.6
1:0.9	+7.6	−11.7	−5.5	+8.5	−4.4	−8.1	+4.4	+9.1	+0.1	−14.8	−1.0	+4.3
1:0.8	**+29.8**	−15.3	+5.5	−9.5	+20.4	**+45.1**	−2.4	−1.9	−3.1	−20.3	−5.4	+4.7
1:0.7	+19.2	−12.8	+2.7	−11.6	+11.8	−18.6	+0.7	−5.4	+11.1	+7.3	−9.8	−17.9
1:0.6	−61.1	−54.0	−57.7	−40.4	+2.1	−18.6	−63.9	−53.4	−4.5	−34.2	−57.1	−52.7
1:0.5	−50.6	−32.3	−36.4	−56.8	−11.0	−17.6	−55.7	−54.3	−29.2	−54.0	−56.8	−52.0
1:0.4	−12.4	+3.9	−43.1	−53.1	−52.7	−50.2	−57.0	−63.8	−36.6	−52.4	−78.2	−76.2
1:0.3	−65.0	−44.5	−78.3	−57.4	**+173.7**	**+274.7**	−81.2	−78.8	−30.5	−58.6	−79.5	−79.4
1:0.2	−79.1	−76.8	−80.8	−78.9	−66.5	−66.2	−86.3	−87.4	−22.2	−45.8	−91.2	−90.3

Three observations sharpen the diagnosis.

Injection at mild coarsening is systematic, not exceptional. The implicit-clamped pairs (II, IE) inject at nearly every overlap for ratios 1:0.9–1:0.7: II +29.8% at max/1:0.8, IE +45.1% at $\frac{1}{2}$/1:0.8, and seven further cells above +7%. This is with the variational (contraction) transfer already in place: the contraction property bounds the *transfer* but not the sign of the dashpot work, confirming that no per-side transfer improvement yields sign-definiteness on the two-surface overlap coupling (cf. the reframed Phase 1).

A violent injection resonance. At $\frac{1}{2}$/1:0.3 the coupling pumps the beam to +174% (II) and +275% (IE) of $E(0)$. The energy grows smoothly for ∼0.7 ms and saturates near 3–4 × $E(0)$ — steady interface pumping, not a numerical explosion. At this ratio the clamped mesh has degenerated to a single-element cross-section (seven bricks in total; one element through the thickness from 1:0.7 down), so it cannot represent the bending wave at all. Degeneracy alone does not decide the outcome, however: the *same* clamped mesh at quarter overlap dissipates −30.5% (II), and the explicit-clamped pairs on the identical configuration dissipate −81%. Whether the sign-indefinite interface work pumps or absorbs is a property of the mesh *pair* and the overlap geometry together.

The drift is violently non-monotone in the ratio. At $\frac{1}{2}$ overlap, II moves −52.7 → +173.7 → −66.5% across the three adjacent ratios 1:0.4 → 1:0.3 → 1:0.2. There is no usable regime map "mild contrast = small drift": both the sign and the magnitude of the energy defect are extremely sensitive to the discrete mesh pair.

Practically, this closes the question of whether the nonconforming overlap coupling can be trusted with a drift *budget*: it cannot. The interface work is sign-indefinite (Sec. 2), and this sweep shows both signs realized at large amplitude across neighboring points of a smooth parameter family. Data: `ratio-sweep/` in the study bundle.

4 What the literature says

Four independent lines of analysis (all in `~/Downloads/Optimized-Schwarz-Methods`) agree on the structure of the problem and of its solution.

4.1 The interface-work pairing identity

For Newmark-family subdomains coupled by interface tractions, the discrete energy balance over a step contains the interface term (Gravouil–Combescure 2001, Eq. (34); Karimi–Nakshatrala 2014, Eq. (84))

$$E_{\text{int}} = \sum_{\text{sides } k} \frac{1}{\Delta t_k} [\![\dot{\boldsymbol{u}}_k]\!]^T \boldsymbol{C}_k^T [\![\Lambda_k]\!], \tag{2}$$

the work of the transmitted tractions through the kinematic jump, each side on its own grid. It vanishes iff the traction is a *single-valued* field acting equal-and-opposite on the two sides through *adjoint* trace operators, with velocity continuity enforced at instants shared by both time grids. Gravouil–Combescure prove their subcycled coupling makes $E_{\text{int}} \leq 0$ (dissipative, never a source); Prakash–Hjelmstad restructure the intermediate tractions so it telescopes to exactly zero. Neither treats spatially nonconforming interfaces; both warn that breaking the pairing leaves E_{int} of indefinite sign — precisely the $+554$ J observation.

4.2 Representability: the converged nonconforming solution

Gander–Halpern–Nataf (SINUM 2003) analyze discrete Schwarz waveform relaxation with impedance transmission on nonmatching grids. Their key statements: the choice $Z = \rho c$ of the *partner* side is the exact transparent operator in 1D and optimal-in-class generally (§3, Eq. (3.10)) — so the impedance choice is not the problem; their nonmatching-grid convergence proof requires the intergrid transfer to be an L^2 *contraction* (projection qualifies, Eq. (5.24); pointwise interpolation does not); and, for nonconforming grids, “the coupled system *defines* the solution when converged” (§5.4) — there is no monodomain scheme it equals. Continuity at the converged fixed point holds only *through the projection*: the component of the fine trace orthogonal to the coarse trace space is invisible to the partner and never matched. The impedance dashpot then acts on that persistent component as a genuine absorbing boundary — sign-definite dissipation that is the price of stability, not an iteration defect. The cure they and Achdou–Japhet–Maday–Nataf (Numer. Math. 2002) point to is not a better Z but a *common trace space* for the datum entering the dashpot.

4.3 The DG telescoping condition

Energy-stable discontinuous-Galerkin couplings (Appelö–Wang 2019; Duru et al. 2021; Liu et al. 2023) make the interface energy identity exact by construction: a single “starred”/Riemann state is computed on a common description of the interface (the mesh intersection) and enters *both* sides’ weak forms; the interface power then telescopes to

$$P \;=\; -\int_\Gamma [\![\dot{\boldsymbol{u}}]\!]^T \boldsymbol{Z} [\![\dot{\boldsymbol{u}}]\!] \,\mathrm{d}S \;\le\; 0 \qquad \text{(or exactly 0 for the central variant)}, \tag{3}$$

with the dissipation confined to the jump and vanishing under refinement and at convergence. The precise discrete requirement: if side k receives partner kinematics through $\Pi_{j\to k}$, the transfers must be L^2-adjoints through the variational cross-mass matrix,

$$\boldsymbol{M}_1\Pi_{2\to1} \;=\; \big(\boldsymbol{M}_2\,\Pi_{1\to2}\big)^T \;=\; \boldsymbol{B}, \qquad B_{mn} = \int_\Gamma \varphi^1_m\,\varphi^2_n\,\mathrm{d}S, \tag{4}$$

with $\boldsymbol{M}_k$ the side-k boundary mass matrices and the integrals evaluated on the common refinement. Two independent pointwise interpolations violate (4), the cross terms do not cancel, and the interface term has no sign. Liu et al. additionally warn that nonconforming cross-interpolation matrices are catastrophically ill-conditioned and that interface unknowns may need implicit treatment even between explicit interiors (their IMEX-Newmark) — directly relevant to the conforming IE/EE growth (+5 to +10%), where the explicitly-evaluated dashpot at lagged time levels injects energy that central-difference ($\gamma = \frac{1}{2}$, zero algorithmic damping) never removes.

5 Synthesis: why each observed regime occurs

Conforming, implicit. Transfer is the identity (node-aligned), the adjoint condition (4) holds trivially, the jump closes at Schwarz convergence, the dashpot does no work: conserved.

Nonconforming, dissipative majority. The jump cannot close (representability); the dashpot absorbs the unrepresentable content persistently. Loss grows with mesh ratio (more unrepresentable content) and shrinks with overlap width (the overlap buffers the boundary from the fine content, and a wider overlap gives the coarse subdomain more elements to express the transferred field).

Nonconforming, growth cases. Non-adjoint pairing makes the dashpot term sign-indefinite; wherever the positive branch dominates the coupling pumps instead of absorbing. The ratio sweep (Sec. 3) shows this is generic — systematic injection at mild coarsening, +275% at the worst cell — and that the sign flips between adjacent mesh ratios. Not stochastic — structural.

Conforming, explicit free subdomain (IE/EE). Time-level, not space, pairing failure: the dashpot force is evaluated explicitly at lagged partner velocity; central difference has no algorithmic damping to absorb the error.

Variational rescue. The variational (L^2) projection is a contraction and the characteristic transfer restores much of the traction consistency, but the dashpot still sees the full jump; the outcome rides on near-cancellation of two large channels.

6 Redesign: an energy-consistent impedance overlap

Phase 1 — variational transfer by default *(implemented, reframed).* The full adjoint pairing of (4) presumes a *shared* interface on which the two sides' works telescope. The overlap variant has none: data are exchanged on two distinct surfaces Γ_1, Γ_2, so no pairing of their works exists, and energy consistency can come only from agreement of the two solutions in the overlap — exactly what dispersion mismatch precludes at fixed h. The implementable core of Phase 1 is therefore the contraction requirement: `transfer: variational` is now the *default* for the impedance overlap BC (pointwise remains an explicit legacy opt-in; the two coincide on node-aligned interfaces). Empirically this converts the worst sign-indefinite case (+554 J of positive dashpot work, +12.5% growth) into sign-correct controlled dissipation, and improves 17 of 24 nonconforming cases — but it cannot eliminate the dispersion-jump dissipation, and isolated growth cells persist (max/3:4: +1.1 → +14.6%). True sign-definiteness by adjoint pairing belongs to the *nonoverlap* impedance variant, which does have a shared Γ; implementing (4) there is the natural follow-up for energy-critical nonconforming coupling.

Phase 2 — dashpot on the representable jump *(implemented; falsified).* Replace $\boldsymbol{Z}(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}})$ by $\boldsymbol{Z}\,\widehat{\Pi}\,(\dot{\boldsymbol{u}}_p - \dot{\boldsymbol{u}})$, where $\widehat{\Pi}$ is the W-orthogonal projection onto the transferable span — the projector already built by the content-absorption machinery (`compute_impedance_overlap_schwarz_projectors!`); implemented as `representable dashpot: true` on the BC (a strict no-op on node-aligned interfaces). The premise was that the persistent jump is trace content the coarse space cannot represent, so that filtering it would let it reflect (conserving) instead of being absorbed. *Measurement falsifies the premise*: on the $\frac{1}{4}$/3:4 case the fine side's $\widehat{\Pi}$ is a genuine projection ($\|\widehat{\Pi} - I\|/\sqrt{n} = 0.48$) yet the dashpot work is unchanged (−1066 → −1065 J) — the jump lies *inside* the representable subspace. The persistent jump is therefore not a spatial-representability defect but a *solution-content* defect: coarse-grid dispersion phase-lags the transiting pulse, so at Schwarz convergence the two discrete solutions disagree by a smooth, in-band field no static spatial filter can remove. Where $\widehat{\Pi}$ does bite (max/1:2), per-side filtering worsens the transfer asymmetry and the drift (+12.5 → +45.6%). The option is retained (default off) as a diagnostic of *where* the jump lives, and the dimensional caveat stands: the partner's volume-trace span on a boundary with few nodes is generically onto, so trace-representability is the wrong notion of "what the partner can carry" — the right notion is dynamic (dispersion band), not static.

Phase 3 — harmonic-mean impedance. Use $Z = Z_1 Z_2/(Z_1 + Z_2)$ per wave family (the Riemann-solver weighting of Duru et al. and Liu et al.), reducing to $Z/2$ for identical materials and correct for dissimilar ones; the current one-sided partner impedance over- or under-damps whenever materials or resolved impedances differ.

Phase 4 — implicit interface for explicit subdomains. Evaluate the interface (dashpot) terms implicitly at the synchronization instants even when a subdomain's interior is explicit (Liu's IMEX-Newmark; equivalently Karimi–Nakshatrala's velocity continuity at shared instants), targeting the conforming IE/EE growth.

Regression metric. Keep the channel-resolved interface-work CSV as a test: the dashpot work must be ≈ 0 on conforming meshes at Schwarz tolerance, and the study's drift table is the acceptance benchmark for any future change to the transmission machinery. (A universal dashpot sign test is *not* attainable in the overlap variant — see the reframed Phase 1.)

7 The nonoverlap route: adjoint pairing implemented

The redesign identified the *nonoverlap* impedance variant — one shared interface Γ, transmission $\boldsymbol{t} + Z\dot{\boldsymbol{u}} + \alpha W \boldsymbol{u} = g$ — as the member of the family on which the adjoint condition (4) is actually available. This has now been implemented and measured on the cantilever benchmark, split at $x = 0.127$ m with the same mesh-ratio family as Sec. 3 (free side fixed at $h = 6.35$ mm, clamped side coarsened to $6.35/r$ mm).

7.1 Baseline: the legacy coupling is as bad as the overlap family

The pre-existing nonoverlap impedance coupling (per-side transfer operators, per-side impedance and Robin parameter) loses 7–18% at mild mesh contrast — *including on conforming meshes* (−9.6% at 1:1) — and injects violently at strong contrast (+223% EI and +363% EE at 1:0.3; +77% II at 1:0.2). Probes isolate the conforming loss: it is unchanged under $\alpha \to 0$, under equalized α, and under $\Delta t \to \Delta t/2$ (−9.6% in all cases), with the Schwarz iteration converged to 10^{-8}. It is therefore a *consistency* defect of the fixed point itself: the coupling transfers the partner's *static* reaction $-\boldsymbol{f}^{\text{int}}$, but in dynamics the two sides' static reactions are not equal and opposite — they differ by the interface nodes' inertia, each side carrying its own tributary mass. The coupled fixed point then differs from the monodomain discretization by $O(m_\Gamma\, a)$, measured as a steady, Δt-independent interface drain.

7.2 The paired coupling

Enabling `adjoint pairing: true` on both sides changes three things: (i) the exchanged traction becomes the dynamically consistent d'Alembert reaction $\boldsymbol{M}\boldsymbol{a} + \boldsymbol{f}^{\text{int}} - \boldsymbol{f}^{\text{body}}$, so the two interface rows sum exactly to the monodomain row and the monodomain trajectory is the fixed point; (ii) both sides derive their transfer operators from *one* shared cross-mass matrix B (integrated over the finer side's facets), with $\Pi_1 = W_1^{-1}B$, $\Pi_2 = W_2^{-1}B^T$, and each side's force transfer the adjoint of the partner's kinematic transfer ($N_1 = \Pi_2^T$, $N_2 = \Pi_1^T$), which is exactly (4); (iii) the pair shares one impedance ($2Z_1Z_2/(Z_1{+}Z_2)$, the Riemann weighting) and one Robin α.

On parameter selection: the impedance is *estimated*, the Robin parameter is *chosen*. Each side's $Z_k = \rho_k c_{p,k} = \sqrt{\rho_k(\lambda_k + 2\mu_k)}$ is computed from its own material data, and the pair impedance is their harmonic mean, so the user supplies nothing. The Robin parameter α is user input (`robin parameter`, default 0), and no estimate is attempted because none is needed for consistency: at the Schwarz fixed point the two Robin conditions add to traction balance and subtract to $Z\,\dot{\boldsymbol{\delta}} + \alpha W \boldsymbol{\delta} = 0$ for the interface jump $\boldsymbol{\delta}$, so with $\boldsymbol{\delta}(0) = 0$ the converged solution is independent of $Z > 0$ and $\alpha \geq 0$ — both act only on the convergence *rate* of the iteration (the optimized-Schwarz role of the zeroth-order Robin term) and on the interface energy bookkeeping en route. What consistency does require is that α be *one* number per interface: the Robin spring stores and returns energy conservatively only when both sides' interface forces pair through the same coefficient. Mismatched values under pairing are therefore a hard input error — Norma aborts, naming both side sets and values — rather than a warning with an internal average, since silent averaging would apply a condition neither side stated. Drift at 1 ms, implicit–implicit:

ratio	legacy II	paired II
1:1.0	−9.61	**−0.03**
1:0.9	−6.84	**−0.03**
1:0.8	−7.96	−3.59
1:0.7	−2.06	−6.55
1:0.6	−9.39	−9.71
1:0.5	−13.25	−13.32
1:0.4	−45.78	−15.13
1:0.3	+18.95	−15.08
1:0.2	+77.49	−13.26

Three properties, in order of importance. First, *no injection at any ratio*: the interface work is sign-definite, as the telescoping identity (3) guarantees — the wild −53 → +174 → −66% swings of the overlap family and the +363% blowups of the legacy nonoverlap coupling are gone. Second, near-conservation through mild contrast (−0.03% at 1:0.9, where every other variant tested loses or gains several percent). Third, the residual dissipation at strong contrast is the *genuine* trace-space jump being absorbed at the rate $Z\int_\Gamma [\![\dot{\boldsymbol{u}}]\!]^2$: at the exactly nested ratio 1:0.5 (where the legacy transfer is coincidentally adjoint and quadrature is exact) paired and legacy agree to 0.1%, so the remaining loss is not a transfer artifact but the Gander–Halpern–Nataf price of coupling trace spaces that genuinely cannot represent each other's content — now paid in a controlled, monotone, sign-definite form.

7.3 Explicit subdomains: IMEX interface treatment

With an explicit (central-difference) subdomain on either side, the paired coupling is violently unstable ($+10^3$ to $+10^5$ %) if the interface terms are evaluated at lagged time levels: the consistent traction couples the partner's *acceleration* into the interface force, an algebraic loop that the implicit pairs resolve within the Schwarz iteration. The cure is the Phase 4 structure (Liu et al.'s IMEX-Newmark), now implemented: because the dashpot force is linear in the unknown acceleration, the explicit update is closed implicitly on the interface rows only,

$$\big(\boldsymbol{M} + \gamma\,\Delta t\,Z\,W\big)\big|_\Gamma\,\boldsymbol{a}_{n+1} \;=\; \boldsymbol{M}\,\boldsymbol{a}^{\mathrm{expl}}\big|_\Gamma, \tag{5}$$

a small SPD solve per interface (reused for the three components) while the interior stays matrix-free explicit; the dashpot then acts on the end-of-step velocity $\dot{\boldsymbol{u}}_{n+1} = \tilde{\boldsymbol{u}} + \gamma\Delta t\,\boldsymbol{a}_{n+1}$ and the Schwarz iteration synchronizes both sides at t_{n+1} exactly as in the implicit–implicit case. Paired drift at 1 ms, all combos:

ratio	II	IE	EI	EE
1:1.0	−0.03	−10.3	−4.1	−8.5
1:0.9	−0.03	−8.2	−4.5	−9.4
1:0.8	−3.6	−11.6	−10.2	−16.0
1:0.7	−6.6	−15.9	−12.2	−19.0
1:0.6	−9.7	−19.2	−25.0	−27.3
1:0.5	−13.3	−21.2	−21.0	−24.9
1:0.4	−15.1	−25.6	−21.3	−29.4
1:0.3	−15.1	−21.6	−33.8	−37.2
1:0.2	−13.3	−30.2	−30.0	−35.3

Sign-definite everywhere: no injection at any ratio in any combination. The explicit columns of this table must, however, be read against the right reference: a *monodomain* central-difference run of the same beam reads −8.2% in the energy metric used here (consistent-mass kinetic energy sampled at integer steps). Against that reference the conforming paired EE run is exact to 0.35% *pointwise over the whole millisecond* — the coupled trajectory *is* the monodomain trajectory, as the fixed-point argument predicts.

Resolved: the −8.2% band was the metric, not the physics. Central difference advances the *lumped*-mass system with leapfrog structure, so the discrete energy it conserves exactly (linear problems) is the staggered lumped-mass form

$$E_n \;=\; \tfrac{1}{2}\,\dot{\boldsymbol{u}}^T_{n-\frac{1}{2}}\boldsymbol{M}_L\,\dot{\boldsymbol{u}}_{n+\frac{1}{2}} + \mathrm{SE}(\boldsymbol{u}_n) \;=\; \tfrac{1}{2}\,\dot{\boldsymbol{u}}^T_n\boldsymbol{M}_L\,\dot{\boldsymbol{u}}_n - \tfrac{\Delta t^2}{8}\,\boldsymbol{a}^T_n\boldsymbol{M}_L\,\boldsymbol{a}_n + \mathrm{SE}(\boldsymbol{u}_n). \tag{6}$$

Evaluating instead the consistent-mass integer-step kinetic energy misprices the short-wavelength content of the sharp release front twice over (consistent vs. lumped Rayleigh quotients diverge at high wavenumber, and the integer-step velocity aliases the staggered content). The staggered correction alone is Δt-dependent and small; the mass operator is the dominant term — which is why the band originally read as "Δt-independent mesh dispersion." With the energy CSV evaluating (6) for central-difference subdomains (implemented in `overlap_energy.jl`), the monodomain explicit run conserves to $\pm 0.000\%$ over the full millisecond, and the paired table becomes, at 1 ms:

ratio	II	IE	EI	EE
1:1.0	−0.03	−0.11	+0.20	−0.27
1:0.9	−0.03	−0.10	+0.29	−0.27
1:0.8	−3.6	−4.4	−3.5	−7.3
1:0.7	−6.6	−6.2	−5.3	−9.8
1:0.6	−9.7	−9.8	−9.8	−11.8
1:0.5	−13.3	−13.8	−10.2	−10.5
1:0.4	−15.1	−17.8	−11.4	−13.3
1:0.3	−15.1	−11.5	−26.4	−20.2
1:0.2	−13.3	−20.1	−27.8	−21.4

On conforming and mildly nonconforming meshes *all four* integrator combinations now conserve to $\leq 0.3\%$; the nonconforming columns carry the same monotone, sign-definite trace-jump dissipation in every combination. (The tiny positive EI entries, +0.2–0.3%, are bounded transient storage in the conservative interface spring, not secular growth.)

7.4 Subcycled (multi-time-step) pairing

Different time steps per subdomain require no new exchange machinery: each subdomain records its per-substep trajectory (kinematics *and* internal force) during every Schwarz iteration, the coupling evaluates the partner's state in time at each of its own substeps, and the paired d'Alembert traction follows. (Enabling this exposed a time-step erosion bug in the subcycler — the stop-aligned remainder step persisted as the nominal step and the Schwarz re-integration amplified the floating-point remainder geometrically until the step collapsed — fixed independently of the pairing.)

The exchange was *intended* to interpolate the partner piecewise-linearly in time — the interpolated-continuity structure under which Gravouil–Combescure prove the subcycled interface work dissipative ($E_{\text{int}} \leq 0$, $O(\Delta t)$) — but two infrastructure defects, found while implementing the Prakash–Hjelmstad refinement below, meant that is not what actually ran. First, the history interpolation selected the wrong segment (an off-by-one that returned the end-of-history value for any query in the last segment, and extrapolated backward off the following segment elsewhere). Second, the per-stop history was anchored at the *first substep*, not the stop start, so a non-subcycling partner's "history" was a single end-of-stop entry. Net effect: both directions of the exchange end-sampled the partner's stop-end state — constant, not interpolated, data. With both defects fixed the exchange is genuinely piecewise-linear in both directions. Drift at 1 ms, free side subcycled 4:1 ($\Delta t = 0.25\,\mu$s inside the $1\,\mu$s stop, clamped side at $1\,\mu$s; measured on one harness for all columns):

case	end-sampled (as shipped)	interpolated (GC, fixed)
conforming II	−2.8	−3.6
conforming IE	−9.6	−10.1
conforming EE	−10.6	−9.6
1:0.5 II	−20.6	**−14.1**
1:0.5 IE	−24.4	**−21.3**

(Both columns are measured in the consistent-mass integer-step metric for comparability with the development history; in the staggered lumped-mass metric of (6) the shipped interpolated exchange reads conforming II −3.6, IE −1.4, EE −1.2, and 1:0.5 II −14.1, IE −14.3% — the explicit subcycled overhead is about one percent on conforming meshes.) Genuine interpolation wins decisively at mesh contrast, where end-sampling aliases the fine side's intra-window motion into the exchanged traction (−20.6 → −14.1%). It *loses* slightly on the conforming implicit cases because the end-sampled datum is not merely aliased — being the stop-*end* value applied throughout the window, it *leads* the true trajectory, and that anticipatory bias is a mild energy-injection channel that happened to offset part of the GC dissipation. A cancellation between two discretization crimes is not a conservation property, and it collapses exactly where the stakes are high (strong contrast); the interpolated exchange is now the behavior of the paired coupling.

The Prakash–Hjelmstad refinement is not partitioned-compatible

Prakash–Hjelmstad telescope the subcycled interface work to exactly zero by giving the interface traction a linear-in-time representation $\hat{\lambda}(t)$ whose step values satisfy $\frac{1}{2}(\hat{\lambda}_n + \hat{\lambda}_{n+1}) = \overline{\lambda}_{n\to n+1}$ (the partner's window average), so the trapezoidal work sums of the two sides cancel window by window. Three Schwarz-compatible constructions of this datum were implemented and measured on the benchmark above:

Window average. Handing the coarse side the partner's trapezoidal window average is work-conjugate to the fine side's interpolation — but an average estimates the window *midpoint*, and the half-window lag of the interface force behind the interface motion is artificial damping: uniformly worse than plain interpolation (−3.6 → −4.2, −10.1 → −11.1, −9.6 → −12.3, −14.1 → −15.5, −21.3 → −23.0).

Exact telescoping recursion. The de-lagged datum $\hat{\lambda}_{n+1} = 2\overline{\lambda} - \hat{\lambda}_n$ satisfies the telescoping identity exactly — and diverges: the factor of two on the partner-state channel doubles the Schwarz fixed-point iteration's contraction factor, and the coupled iteration blows up within tens of stops.

Least-squares endpoint fit. The endpoint value of the LS linear fit of the partner trajectory over the window is lag-free, filters intra-window content instead of aliasing it, and telescopes for window-linear tractions. It is stable over tens of stops and passes the short regression — and over hundreds of stops it pumps energy catastrophically ($+10^3$–10^5% before aborting): a zero-lag filtered estimate is an extrapolation, and its frequency response exceeds unity off the linear band.

The pattern is the classical stability folklore of partitioned coupling: only *causal* transfer operators (lag ≥ 0) are robust in a fixed-point exchange, and every zero-lag work-conjugate datum is anticipatory. Prakash–Hjelmstad's telescoping is consistent with this — their method solves the coupled interface problem *directly* (a FETI-style interface system per large step), which is precisely what a partitioned Schwarz sweep does not do. Exact subcycled interface-work telescoping therefore requires a direct interface solve, not a smarter exchange; within the Schwarz family, genuinely interpolated GC transfer is the measured stable optimum, and its $O(\Delta t)$ dissipation (-3.6% at conforming II vs. -0.03% same-step) is the price of subcycling.

8 Where this lands

For wave-transit problems on *nonconforming overlapping* decompositions, the tested coupling family is absorbing-or-unstable: hard Dirichlet matching (pointwise or weak) is violently unstable, and the impedance condition is stable precisely because its dashpot continually absorbs the inter-grid disagreement, whose dominant component is coarse-grid dispersion — a smooth, in-band solution defect that neither spatial filtering (Phase 2, falsified) nor transfer improvements (Phase 1) can close at fixed h. The ratio sweep (Sec. 3) removes the remaining comfort: the interface work is sign-indefinite in practice as well as in principle, so the "price" is not a bounded, predictable dissipation — neighboring mesh ratios swing from -53% to $+174\%$ to -66%. Mesh conformity therefore remains the only route to energy conservation *within the overlap family*; nonconforming overlaps are usable only where neither energy loss *nor* energy gain of tens of percent matters, with the channel-resolved instrumentation reporting which one occurred.

For energy-critical nonconforming coupling, the theory-backed path — the *nonoverlap* impedance variant, whose single shared interface admits the adjoint pairing (4) — is now implemented and validated (Sec. 7): with the consistent d'Alembert traction and the paired operators, implicit–implicit coupling conserves to -0.03% through mild mesh contrast and degrades into monotone, sign-definite, bounded dissipation at strong contrast, with no injection anywhere. Explicit subdomains are covered by the IMEX interface treatment (Sec. 7.3): the conforming explicit–explicit trajectory matches the monodomain explicit run to 0.35% pointwise, and every combination is sign-definite at every ratio. The practical recommendation is therefore: conforming meshes within the overlap family, or the adjoint-paired nonoverlap variant (now the default for `Schwarz impedance nonoverlap`; `adjoint pairing: false` restores the legacy transfer) when the decomposition must be nonconforming and the energy budget matters. Subcycled (multi-time-step) pairing is validated with genuinely interpolated Gravouil–Combescure transfer — two infrastructure defects that had silently reduced it to end-sampling are fixed, halving the excess dissipation at strong mesh contrast (Sec. 7.4). The Prakash–Hjelmstad exact-telescoping refinement was implemented in three Schwarz-compatible forms and measured to be structurally incompatible with a partitioned exchange (lagged forms over-dissipate; zero-lag forms are anticipatory and unstable); realizing it requires a direct FETI-style interface solve per large step, which, together with the localized-Lagrange-multiplier frame construction (Park–Felippa; Dvorak et al. 2023), remains follow-up work.